SLE-3: Architectural Design (Full C4 Model)

Course: 02AML204 – Introduction to Artificial Intelligence

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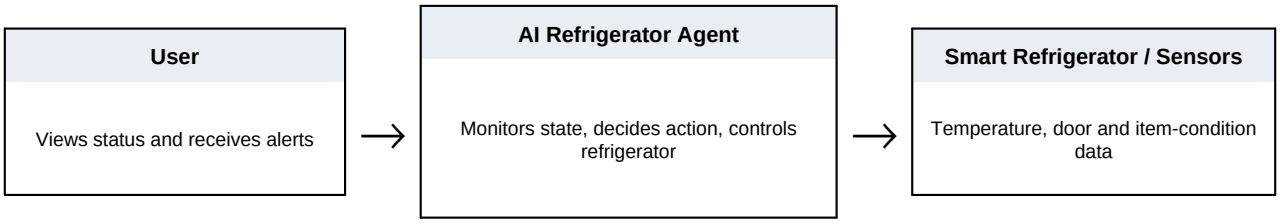
Date: 04-10-2026 **System:** AI Refrigerator Agent

1. System Title & Short Description

AI Refrigerator Agent is an intelligent agent designed to monitor and manage the conditions inside a smart refrigerator. It receives information such as temperature, door status and food/item status through sensors or user input. The agent analyses the current state and decides suitable actions, such as cooling, reducing cooling, giving an alert, or notifying the user. The main goal is to maintain suitable storage conditions while reducing unnecessary energy use.

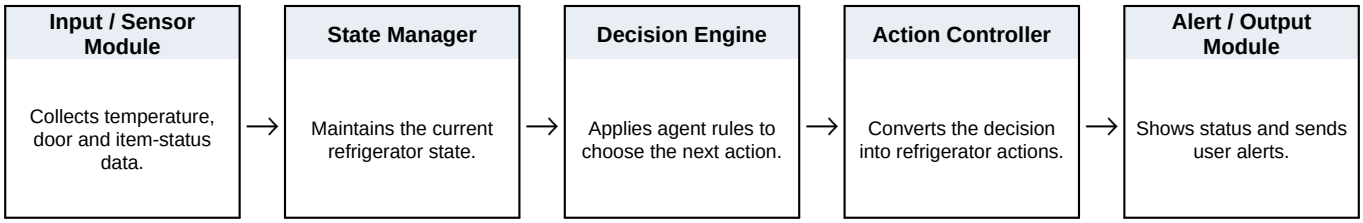
The architecture below follows the four C4 levels required in the SLE-3 guideline: Context, Container, Component and Code. The design is intentionally kept simple so that the complete agent can be understood and explained in a viva.

2. Context Diagram (Level 1)



Context explanation: The User interacts with the AI Refrigerator Agent to see refrigerator status and receive alerts. The refrigerator and its sensors provide the environmental and operational information required by the agent. The agent processes the input and sends decisions or control commands back to the refrigerator, while notifying the user when attention is required.

3. Container Diagram (Level 2)



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3. Container Diagram (Level 2) — Explanation

Input / Sensor Module: Collects temperature, door and item-status data.

State Manager: Maintains the current refrigerator state.

Decision Engine: Applies agent rules to choose the next action.

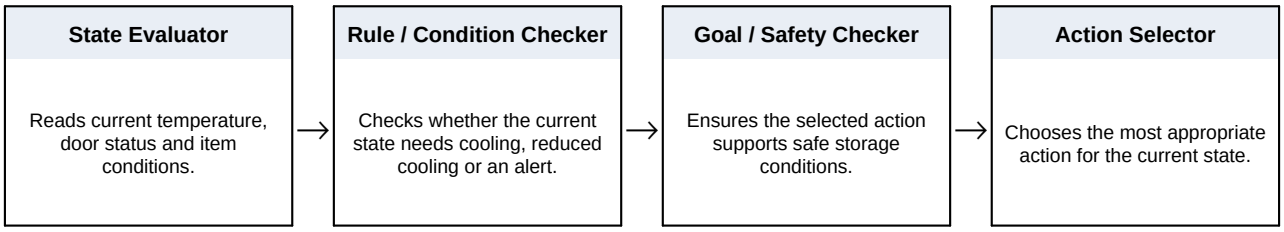
Action Controller: Converts the decision into refrigerator actions.

Alert / Output Module: Shows status and sends user alerts.

The five containers form a simple flow: sensor/user input is collected, the refrigerator state is updated, the decision engine selects an action, the action controller performs or requests that action, and the output module communicates the result. This keeps the architecture within the guideline's recommended 4–7 container range.

4. Component Diagram (Level 3)

Selected main container: Decision Engine



Component explanation: The Decision Engine is the most important container because it converts the refrigerator's current state into an intelligent action. The State Evaluator prepares the relevant state, the Rule/Condition Checker identifies what is happening, the Goal/Safety Checker keeps the decision aligned with safe operation, and the Action Selector produces the final action for the Action Controller.

5. Code Level Overview (Level 4)

Main class / function	Responsibility
class RefrigeratorState	Stores temperature, door status and item-condition information.
class SensorInput	Represents input received from refrigerator sensors.
def update_state()	Updates the current refrigerator state.
def evaluate_conditions()	Checks the current state against operating conditions.
def decide_action()	Selects cooling, reduce-cooling, alert or no-change action.
def execute_action()	Sends the selected action to the refrigerator/controller.

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6. Design Decisions

1. The system is divided into five containers so each responsibility is easy to understand and test. 2. The Decision Engine is separated from input and action handling so the agent's decision-making logic remains clear. 3. A separate State Manager is used because decisions should be based on the refrigerator's current state rather than isolated sensor readings. 4. The design keeps the architecture small and readable, as required by the SLE-3 guideline.

7. AI Contribution Note

AI tools used: ChatGPT.

What AI helped with: Structuring the C4 architecture, organizing the report sections, preparing concise explanations, and checking that the four required C4 levels are represented.

What I did myself: Selected the AI Refrigerator Agent as the system, reviewed the architecture, decided the main containers/components, and will explain the design and agent behaviour during evaluation.

8. Conclusion

This SLE-3 exercise helped in understanding how an AI agent can be represented as a complete software architecture. The AI Refrigerator Agent was described at Context, Container, Component and Code levels. The C4 model makes it easier to move from the overall system view to the main modules, internal decision-making parts and finally the important classes/functions. The architecture is kept simple so that the system structure and design decisions can be clearly explained.